Piyush Yadav

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Experience

Enest-Empowering Tomorrow May 2026 – Aug 2026
Software Developer Intern *Remote*

- Built and maintained 10+ REST API endpoints using Node.js, Express.js, TypeScript, and MongoDB for user management, authentication, and Industry Connect features.
- Designed candidate discovery system with search, filtering, pagination, and aggregation across 3 data models (User, Student, Industry Connect), cutting query complexity.
- Implemented JWT auth with server-side API proxying across Next.js and Express; resolved 3+ production bugs in auth flows, API routing, and MongoDB.
- Audited payment workflows, identified verification gaps, and integrated SendGrid for transactional email delivery (welcome + receipts).

Kyverno Jun 2026 – Present
Open Source Contributor | CNCF, 8k+ GitHub Stars *Remote*

- Merged 8 PRs into Kyverno (CNCF Kubernetes policy engine) fixing policy evaluation, webhook reconciliation loops, namespace selector propagation, and image digest verification in Go.
- Debugged CEL-based policy evaluation and controller reconciliation in Go; added Chainsaw unit tests for ValidatingPolicy and ImageValidatingPolicy edge cases.
- Resolved admission controller webhook, policy autogeneration, and image verification bugs via code review with Kyverno maintainers.

Projects

k6e | *Go, Docker, SQLite, REST, Distributed Systems* | *Read Here*

- Building a lightweight multi-node container orchestration platform with a control plane, worker agents, scheduler, execution layer, persistent SQLite state, and Docker runtime abstraction.
- Implemented Kubernetes-style desired-state reconciliation with replica tracking, liveness detection, heartbeat-based node health, failure recovery, scale-up/down decisions, and idempotent controller behavior across **180+ test cases**.
- Designed assignment and execution workflows with template hashing, explicit lifecycle states, runtime observation, and fault propagation to handle divergence between control-plane state and container state.

DevWrap | *Go, Next.js, Redis, GraphQL, Docker*

- Built an asynchronous GitHub developer analytics platform that transforms repository activity into a shareable dashboard covering contributions, commits, pull requests, reviews, issues, languages, and streaks.
- Architected a decoupled **Next.js → Redis queue → Go worker → GitHub GraphQL → Redis** pipeline with distributed audit locking, background job processing, cached results, and adaptive status polling.
- Implemented at-least-once job handling with Redis processing recovery and containerized the web, worker, and Redis services using Docker Compose.

Uptor | *TypeScript, Node.js, Express, MongoDB, Next.js*

- Built a full-stack observability platform with JWT auth, scheduled health checks, incident tracking, and automated alert state transitions (healthy → degraded → down) for SLA-aware uptime monitoring.
- Architected API, worker, and service layers with concurrency-limited checks and circuit-breaker-style failure handling; resolved SameSite/CORS cookie conflicts in split-domain Vercel + Render deployment.

API Test Suite | *TypeScript, Node.js, Zod, TurboRepo, YAML*

- Built a CLI API testing engine with YAML-defined suites, parallel execution, configurable concurrency, per-request retries, and AbortSignal timeouts.
- Implemented deep validation for HTTP status, body schema (Zod), and latency thresholds; added env-variable templating and structured JSON output for CI/CD integration.

Technical Skills

Languages: Go, TypeScript, JavaScript, Python, C
Backend: Node.js, Express.js, REST APIs, Distributed Systems, Concurrency, RBAC, Background Workers
Databases: MongoDB, PostgreSQL, MySQL, SQLite, Mongoose
Cloud & DevOps: Docker, Kubernetes, Linux, Git, GitHub, Vercel, Render
Tools & Frameworks: Next.js, TurboRepo, Zod, Postman, Selenium, BeautifulSoup
Testing: Chainsaw, Unit Testing, API Testing, YAML-based Test Suites
Open Source: Kyverno, CNCF, Kubernetes Policy Engine, CEL, Admission Controllers, Webhook Reconciliation, PolicyReports

Education

Kamla Nehru Institute of Technology, Sultanpur Expected 2028
B.Tech in Electronics Engineering (GPA: 8.47 / 10.00) *Sultanpur, Uttar Pradesh*

- **Leadership:** Training & Placement Representative (TPR); Secretary of KNIT Sultanpur, Alumni Association